

EE 789: Mid-sem exam/assignment: using RTL

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1 A switch

An NxM switch provides functionality to switch data from 1 of the N input ports to 1 of the M outputs. Let the input ports be numbered 0,1,2,..., N-1 and the output ports be numbered 0,1,2,...,M-1. Input packets arrive at the switch. The packet length is a multiple of 4-byte quantities (words). The first word in the packet is called the *header*, and its format is as follows:

Bits	Interpretation
[31:24]	Destination
[23:8]	Length
[7:0]	Sequence-id

When data enters the switch from a particular input port, the switch examines the header to figure out the destination of the packet. The switch is then responsible for sending the incoming packet to the destination output port in its entirety. The sequence ID identifies the index of the packet (first packet has index 0, second has index 1, etc.).

2 A shared-queue switch

A 2x2 shared-queue switch has the structure shown in Figure 1. Incoming data is kept in a single queue and data from the queue is read out and sent to the appropriate output port.

3 An input-queued switch

A 2x2 input-queued switch has the structure shown in Figure ???. Incoming data is kept in a two input queues and the data from the input queues is sent out to the appropriate output port.

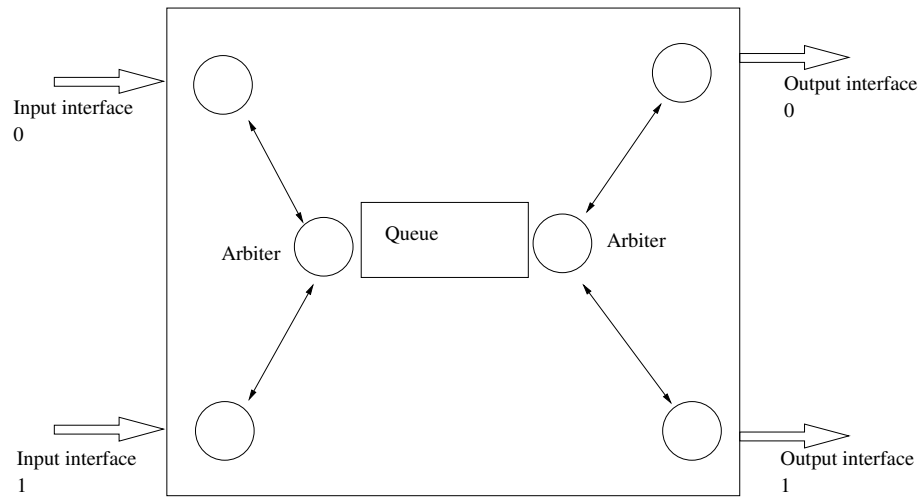


Figure 1: 2x2 Shared Queue Switch

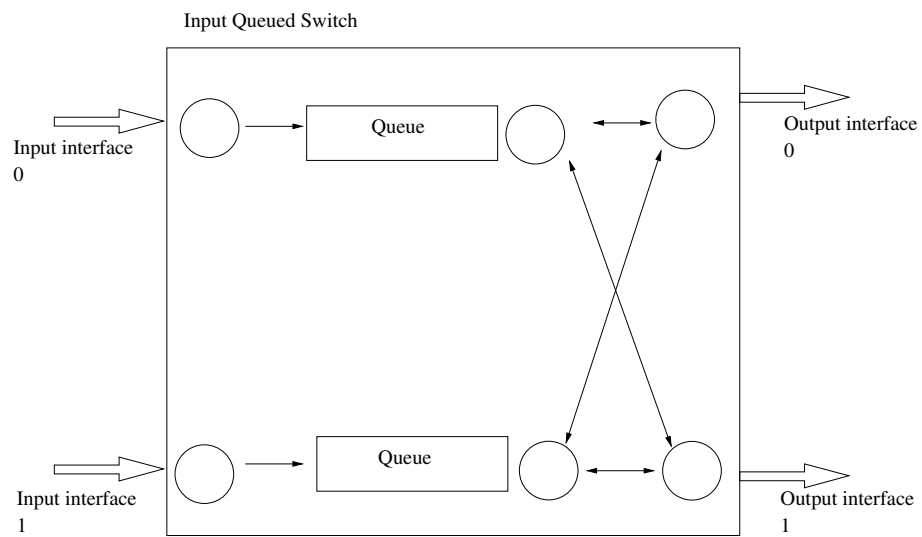


Figure 2: 2x2 Input Queued Switch

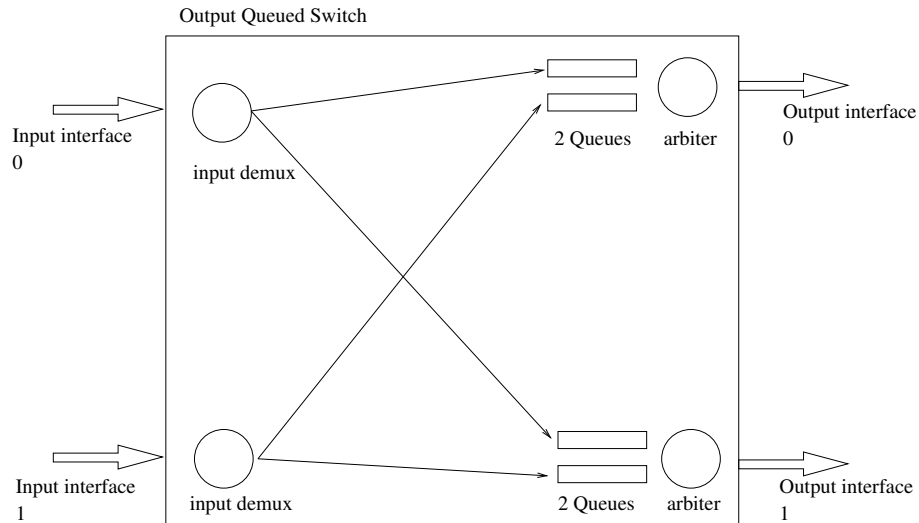


Figure 3: 2x2 Output Queued Switch

4 An output-queued switch

A 2x2 output-queued switch has the structure shown in Figure 3.

Incoming data at an input demultiplexor is written into the appropriate destination queue. Note that if the destination queue has no room, then it will block and the input demultiplexor will have to wait until it is free.

At each output port, there is an arbiter which checks the two queues that it is managing. It serves the queues in a fair manner (that is, in the long term, neither of the two queues will get favoured). The packet in the selected queue is then sent out in its entirety, on to the output port.

5 Assignment

You will design a 2x2 switch, which can handle packets with a maximum size of 256 bytes. You will validate and characterize your switch for the following kinds of traffic:

- Data comes in from a single port and goes out to a single port.
- Data comes in from both ports and goes out to the same destination port.
- Data comes in from both ports and is distributed across the destination ports (incoming packet is equally likely to go to any of the destination ports).

6 How to go about it?

You will first figure out the algorithm, then implement the algorithm in RTL pseudo-code. You will write a test-bench to confirm that the algorithm will function correctly. Please ensure that the test-bench is sufficiently smart so that you can eventually use the same test-bench to characterize the behaviour of the switch for the three kinds of tests outlined above.

7 How will we help you?

We will provide you with a test-bench that handles the third test scenario outlined above. We will also provide you with a VHDL implementation of a queue.